

DA5000W Linear algebra and optimization

October 2025

Tutorial 1

Questions

Question 1:

For the given system of equations,

$$\begin{cases} x + y + z = 2 \\ x + 2y + 3z = 4 \\ x + 3y + 5z = 6 \end{cases}$$

- (a) Check whether this system has a unique solution.
- (b) Use your answer to justify whether the coefficient vectors $(1, 1, 1)$, $(1, 2, 3)$, $(1, 3, 5)$ are linearly dependent or independent.

Question 2:

Find a basis for the subspace defined by the plane

$$x + 2y - 3z - t = 0$$

in $\mathbb{R}^4$. What is the dimension of this subspace?

Question 3:

Let $V = \mathbb{R}^4$, and define the subset

$$W = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + 2x_2 - x_3 = 0, \quad x_4 = 0\}$$

- (a) Prove W is a subspace of V .
- (b) Find a basis and dimension for W .

Question 4:

Define a linear map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by:

$$T(x, y, z) = (x + 2y - z, \quad 3x + y + 4z)$$

- (a) Find the matrix representation of T .
- (b) Find the rank and nullity of T .
- (c) Find a basis for the kernel (nullspace) of T .

Solutions

Solution 1:

- (a) The coefficient matrix is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$$

Calculate its determinant:

$$\begin{aligned} |A| &= 1(2 \cdot 5 - 3 \cdot 3) - 1(1 \cdot 5 - 3 \cdot 1) + 1(1 \cdot 3 - 2 \cdot 1) \\ &= 1(10 - 9) - 1(5 - 3) + 1(3 - 2) \\ &= 1 - 2 + 1 = 0 \end{aligned}$$

Since the determinant is zero, perform row operations for consistency:

$$\text{Row 2: } (1, 2, 3) - (1, 1, 1) = (0, 1, 2)$$

$$\text{Row 3: } (1, 3, 5) - (1, 1, 1) = (0, 2, 4)$$

The lower 2×2 matrix is:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

with determinant zero, indicating dependency.

Check constants difference:

$$4 - 2 = 2, \quad 6 - 2 = 4$$

and ratio $2 : 4 = 1 : 2$ matches coefficients, so system is consistent with infinite solutions, hence no unique solution.

- (b) Since the coefficient matrix has zero determinant and rows are dependent, the vectors are linearly dependent.

Solution 2:

The equation describes a hyperplane; rewrite as:

$$x = -2y + 3z + t$$

Choosing free variables y, z, t and setting one free variable to 1 each time:

- For $y = 1, z = 0, t = 0$: $x = -2$, vector $(-2, 1, 0, 0)$
- For $y = 0, z = 1, t = 0$: $x = 3$, vector $(3, 0, 1, 0)$
- For $y = 0, z = 0, t = 1$: $x = 1$, vector $(1, 0, 0, 1)$

This forms a basis. Dimension = 3.

Solution 3:

- (a) Zero vector $(0, 0, 0, 0)$ satisfies $0 + 0 - 0 = 0$ and $0 = 0$.

For $\mathbf{u}, \mathbf{v} \in W$,

$$(a_1 + 2a_2 - a_3) + (b_1 + 2b_2 - b_3) = 0 + 0 = 0, \quad a_4 + b_4 = 0$$

For scalar c ,

$$c(a_1 + 2a_2 - a_3) = 0, \quad ca_4 = 0$$

So W is a subspace.

- (b) Express variables:

$$x_1 = -2a + b, \quad x_4 = 0, \quad \text{where } x_2 = a, \quad x_3 = b$$

General vector:

$$(-2a + b, a, b, 0)$$

Basis:

$$\{(-2, 1, 0, 0), (1, 0, 1, 0)\}, \quad \dim W = 2$$

Solution 4:

(a) Action on standard basis:

$$T(1, 0, 0) = (1, 3), \quad T(0, 1, 0) = (2, 1), \quad T(0, 0, 1) = (-1, 4)$$

Matrix:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 4 \end{bmatrix}$$

(b) Row reduce:

$$R_2 \rightarrow R_2 - 3R_1 = (0, -5, 7)$$

Rank = 2

Using Rank-Nullity theorem:

$$\text{nullity} = n - \text{rank} = 3 - 2 = 1$$

(c) Kernel satisfies:

$$\begin{cases} x + 2y - z = 0 \\ 3x + y + 4z = 0 \end{cases}$$

Express $x = -2y + z$. Substitute:

$$-6y + 3z + y + 4z = 0 \Rightarrow -5y + 7z = 0 \Rightarrow z = \frac{5}{7}y$$

Then

$$x = -2y + \frac{5}{7}y = -\frac{9}{7}y$$

Kernel basis vector:

$$\{(-9, 7, 5)\}$$